

Socioeconomic & Demographic Drivers of Electric Vehicle Adoption in Washington State

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Problem & Research Question

Goal: Predict EV adoption rate by ZIP code

Why it matters:

- Understand EV adoption patterns
- Support policy and infrastructure planning

Key Research Question: How do median household income, population size, and median age at the ZIP-code level influence the rate of EV adoption (BEV + PHEV) across Washington State?



Data Sources & Preparation

Primary Dataset

WA EV Population Data (2025)

Source: WA Dept of Licensing via [Data.gov](https://data.gov)

276,828 • 16 variables

Key: ZIP Code, Vehicle Type, Make, Electric Range



Secondary Dataset

Census Demographics by ZIP

Source: Data Commons (U.S. Census)

583 ZIPs • 14 variables

Key: Population, Median Age, Median Income

Key Cleaning Step:

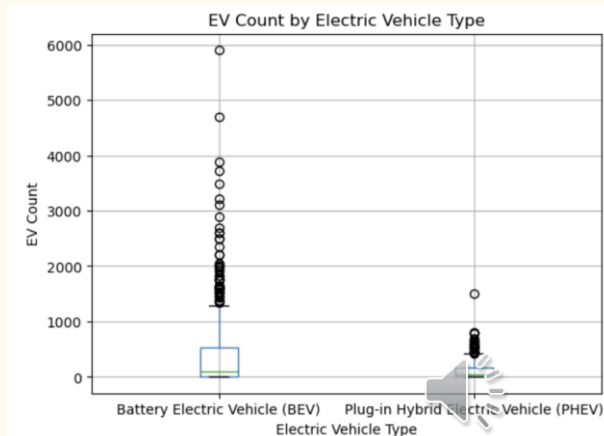
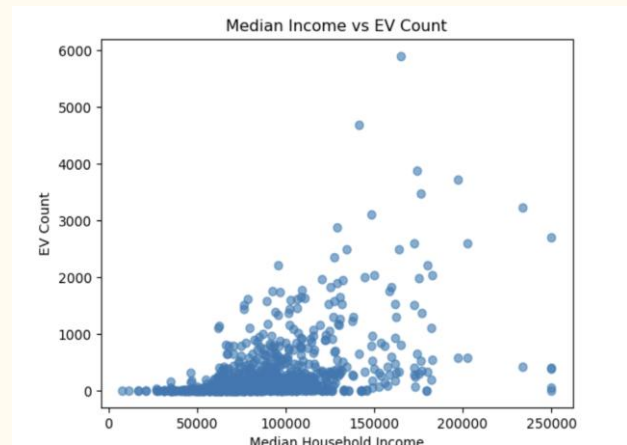
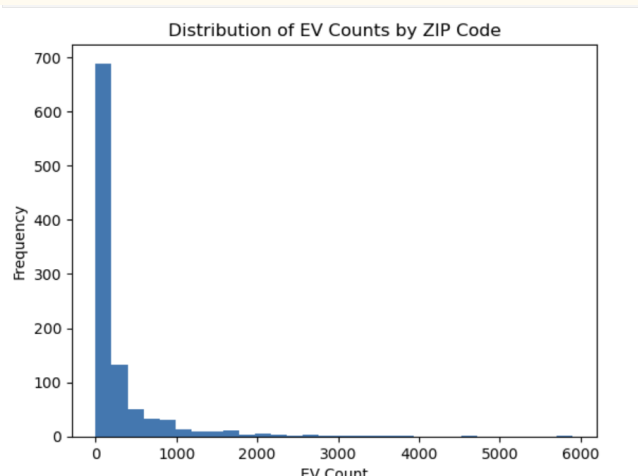
1. Merged datasets on ZIP code (left join — all EV records preserved)
2. Dropped ~9,000 records with missing median household income
3. Standardized column names; corrected data types; masked VIN numbers
4. Computed target variable: $\text{ev_adoption_rate} = \text{ev_count} \div \text{population per ZIP}$



Exploratory Data Analysis (EDA)

Key Finding:

- EV adoption varies by ZIP code
- Higher income areas tend to show higher EV adoption
- Battery Electric Vehicles outnumber PHEVs and show higher median counts per ZIP.



Modeling Approach

Target: $\text{ev_adoption_rate} = \text{EV count} \div \text{population}$ (per ZIP code) | Task: Supervised Regression

Features:

- Median_income_household
- population
- median_age
- zip_code (one-hot encoded)

Linear Regression

Interpretable. Quantifies linear relationship between features and EV adoption. Coefficients show direction + magnitude.

Random Forest

Ensemble method. Captures non-linear patterns and feature interactions. 200 estimators, no max depth.

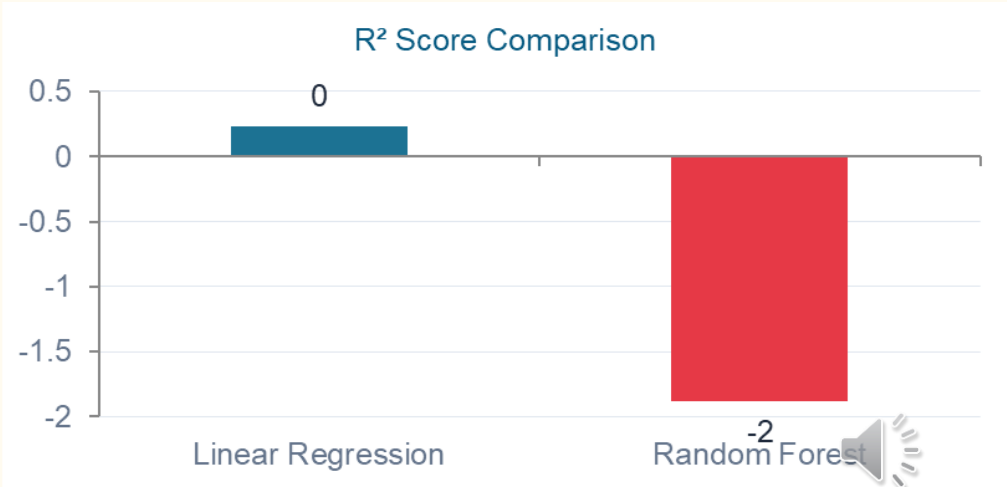


Model Results & Comparison

Model	MAE	RSME	R ²	Verdict
Linear Regression	0.02	0.03	0.236	Selected
Random Forest	0.02	0.05	-1.883	Unstable

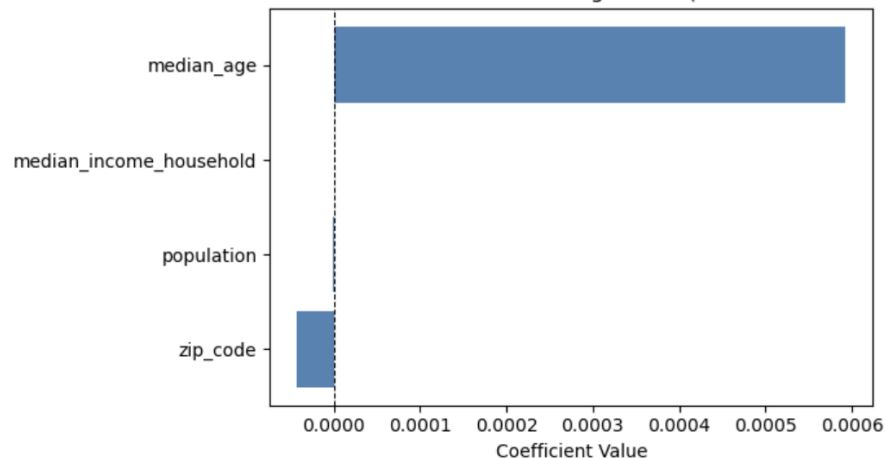
Why Linear Regression Wins

- Positive R² (23.6%) vs negative RF
- More stable cross-validation scores
- Simple, interpretable coefficients
- Random Forest overfits fails on unseen data



Feature Importance & Coefficients

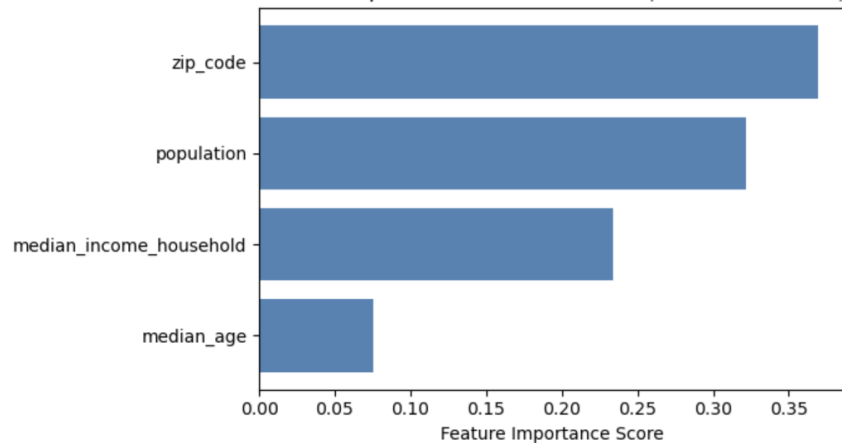
Feature Coefficients - Linear Regression (Numeric Features)



Feature	Coefficient
median_age	5.923729e-04
median_income_household	1.980903e-07
population	-1.805442e-07
zip_code	-4.220761e-05

median_age = dominant predictor in LR

Feature Importance - Random Forest (Numeric Features)



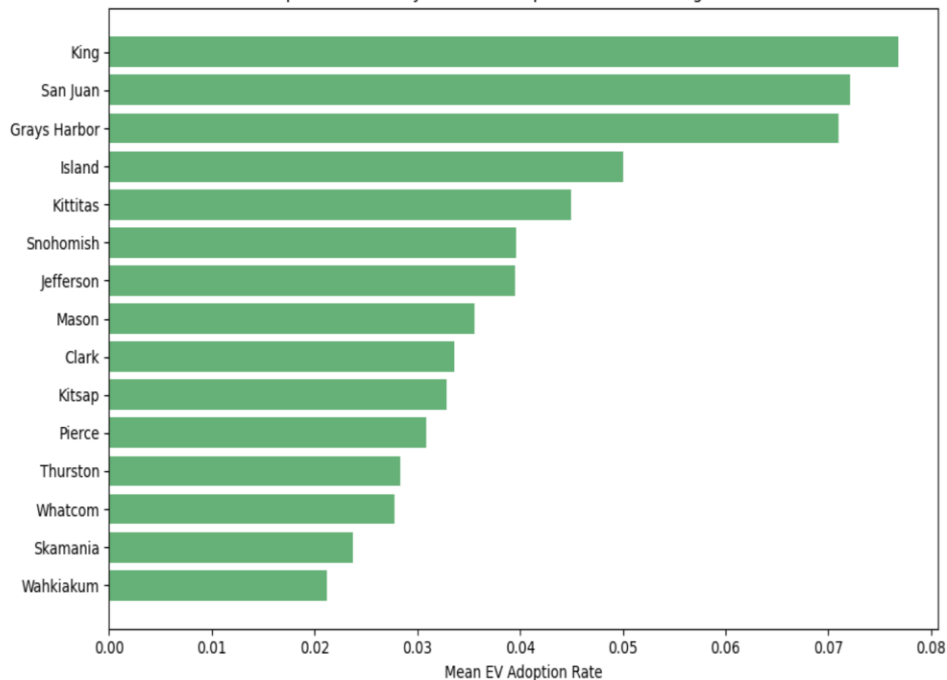
Feature	Importance
zip_code	0.368976
population	0.321792
median_income_household	0.233588
median_age	0.075644

Location (zip_code) + population dominate RF



Geographic Insights — County Level

Top 15 Counties by Mean EV Adoption Rate - Washington State



- King, Snohomish, Pierce dominate the west. Yakima, Franklin, Spokane lag in the east.
- Grays Harbor & San Juan show high adoption despite moderate income — local factors beyond income at play.
- Franklin (0.65%), Pend Oreille (0.66%), Columbia (0.72%) all rural, eastern, low population.
- Income positively correlates with adoption but outliers confirm income alone does not fully explain EV adoption patterns.



Limitations & Ethical Considerations

Limitations

- The model only includes median household zip-code, income, population size, and median age. Many important factors are not included, such as:
 - Housing type (single-family vs. multi-unit housing)
 - Urban density
 - Local EV incentives and policies
 - Public transportation access
 - Total number of cars registered in the same zip codes

Ethical Considerations

- Low adoption rate $\neq$ lack of interest (structural barriers matter)
- Findings show correlation only, not causation
- VIN masked; no personally identifiable info in dataset
- Use findings to improve equity, not penalize communities
- Income predictor may reinforce existing inequalities if misused

